UNIVERSITY OF TEXAS ARLINGTON
COLLEGE OF SCIENCE

Advanced Classical Mechanics

Lecture 3: Matrices & Problem Solving in Mechanics

Readings: Marion - Chapter 1, Morin - Chapter 1

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Matrices

A matrix is a rectangular array of numbers in its most generic definition.

$$\begin{array}{cc} & \begin{array}{cccc} 1 & 2 & \cdots & n \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ \vdots \\ m \end{array} & \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ a_{31} & a_{32} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \end{array}$$

Basic matrix definitions.

1. A Matrix with a **single row** is called a **row vector**.
2. A Matrix with a **single column** is called a **column vector**.
3. A Matrix with the same number of rows and columns is called a **square matrix**.
4. A matrix with no rows or no columns is called an **empty matrix**.
5. The **matrix rank** of is the dimension of the space formed by the independent column vectors of a matrix.

Basic matrix operations.

- **Addition.** The sum $\mathbf{A} + \mathbf{B}$ of two $m \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$ is calculated element-wise:

$$(\mathbf{A}+\mathbf{B})_{i,j} = \mathbf{A}_{i,j} + \mathbf{B}_{i,j}, \quad 1 \leq i \leq m, \quad 1 \leq j \leq n.$$

- **Scalar multiplication.**

$$(c\mathbf{A})_{i,j} = c \cdot \mathbf{A}_{i,j}$$

- **Subtraction.**

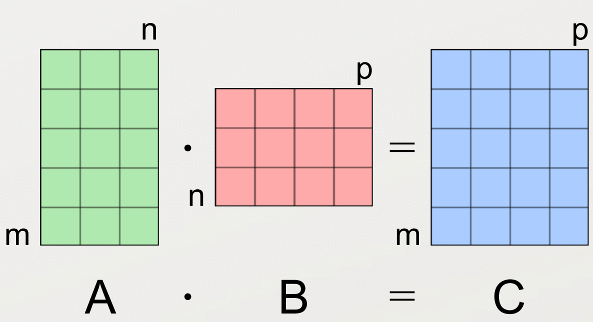
$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-1) \cdot \mathbf{B}$$

- **Transposition.**

$$\left(\mathbf{A}^{\mathrm{T}}\right)_{i,j} = \mathbf{A}_{j,i}$$

- **Product of two matrices ($\mathbf{C} = \mathbf{AB}$).**

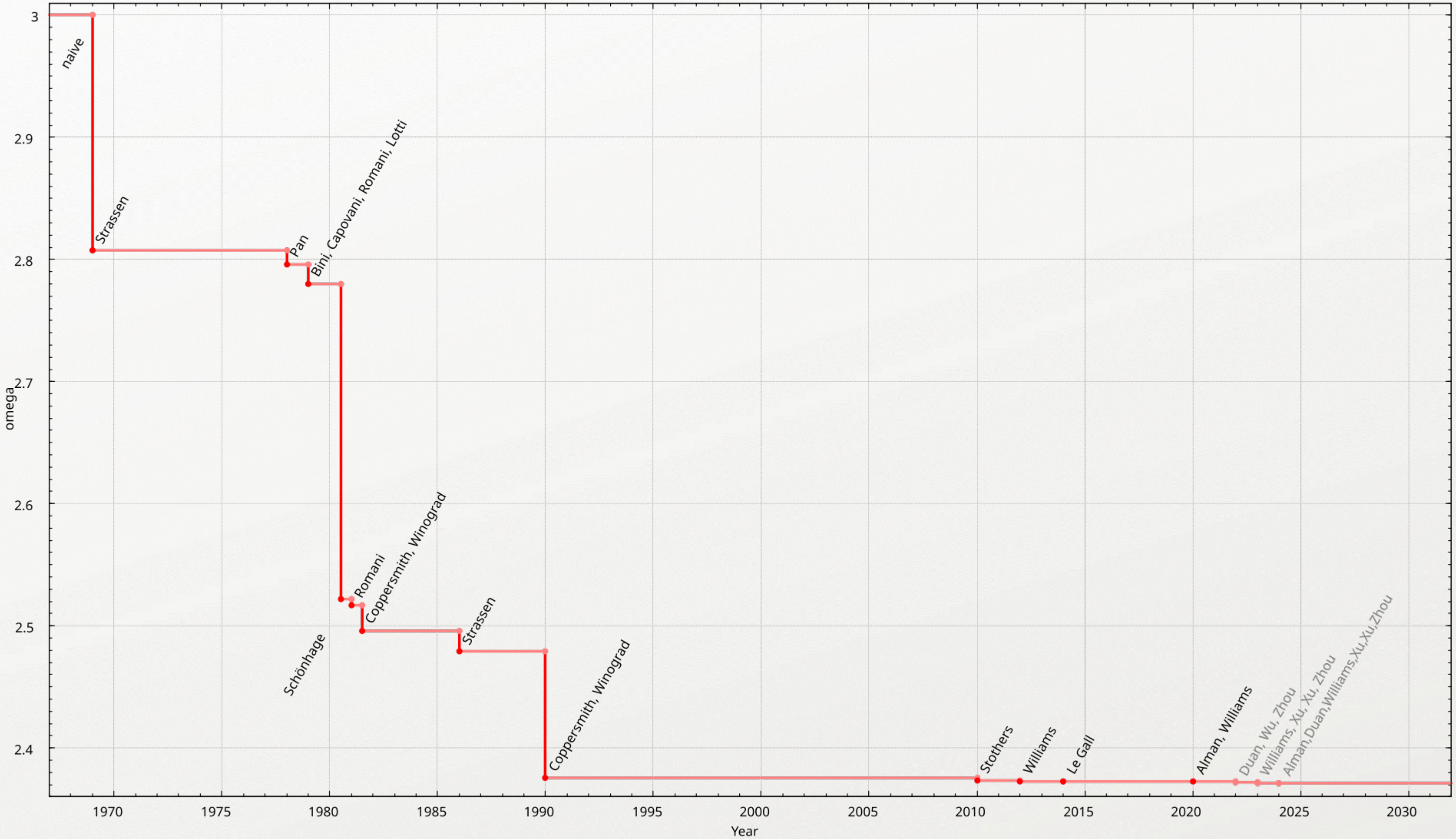
$$[\mathbf{AB}]_{i,j} = a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \cdots + a_{i,n}b_{n,j} = \sum_{r=1}^{n} a_{i,r}b_{r,j},$$



Problem 0. How many scalar multiplications are generally involved in a square matrix multiplication? Could it be reduced?

Matrices

Improvement of estimates of exponent ω over time for the computational complexity of matrix multiplication.



Matrices

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  $$\left(\mathbf{A}^{\mathrm{T}}\right)_{i,j} = \mathbf{A}_{j,i}$$

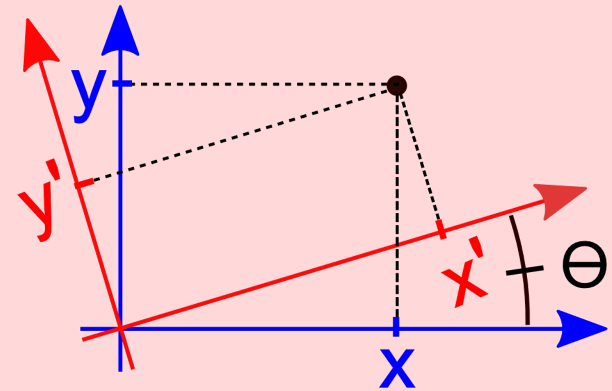
- **Product of two matrices ($\mathbf{C} = \mathbf{AB}$).**
  - In general, matrix multiplication is **non-commutative**: $\mathbf{AB} \neq \mathbf{BA}$
    - Exceptions: Identity matrix: $\mathbf{1A} = \mathbf{A1} = \mathbf{A}$, Inverse matrix: $\mathbf{AA^{-1}} = \mathbf{A^{-1}A}$
  - Matrix multiplication is **associative:** $\mathbf{A}(\mathbf{BC}) = (\boldsymbol{AB})\boldsymbol{C}$

Matrices and coordinate transformations

Matrices are critical for understanding **linear transformations** (a.k.a. **linear maps**).
So what? Linear and other transformations are one of the cornerstones of classical mechanics.

Problem 1. 2D Coordinate system rotation.
Consider the 2D Cartesian coordinate system X-Y.
The frame is rotated counterclockwise at an angle θ, as shown.
Show that for a given arbitrary point, as illustrated,
the following coordinate transformation relationship holds:

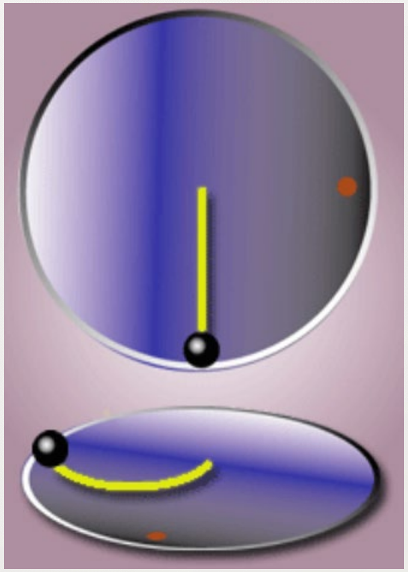
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \quad ,$$



(**Hint:** We have reviewed the polar coordinates and the trigonometric relations in previous lectures and homework. Use the appropriate relations to express (x, y) and (x', y') in polar coordinates. The rest follows.)

How is coordinate transformation related to Classical Mechanics?

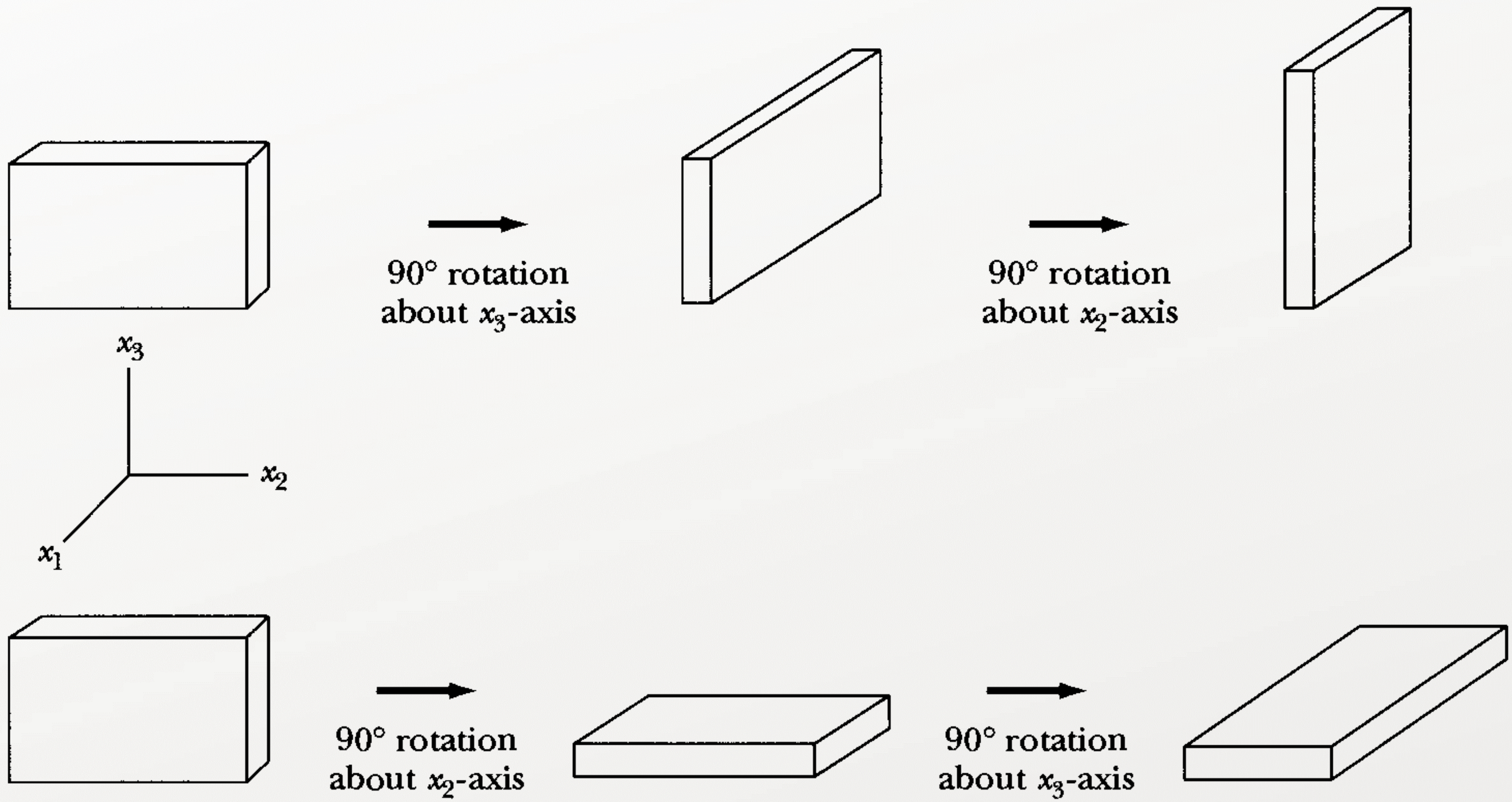
Consider a rotating table as illustrated.
The black ball moves straight in the inertial frame of reference (top figure).
However, the rotating observer (red dot), standing in the rotating/non-inertial frame of reference (bottom figure), sees the object following a curved path due to the Coriolis and centrifugal forces in this frame.



We will get back to this problem later in the semester.

Matrices and coordinate transformations

Beware that the order of rotations matters.



Problem-Solving Strategies in Classical Mechanics

Understanding a problem clearly and entirely is 80% of the effort toward its solution.

- Read the problem statement carefully multiple times, at least twice.
- List all quantitative information given to you in the problem.
- List all the unknown(s) desired and to be obtained in the problem.
- Draw visualizations or diagrams as many as necessary and helpful.
- Think of visualizations, patterns, or equations that connect the knowns to unknowns.
- Upon finding the unknowns or the desired answer, check if the results make sense.

    - Perform as much of calculations as possible symbolically:

        - It is faster.
        - There is less room to make mistakes.
        - You can generate the answer to the problem for all possible input values.
        - It allows you to validate your results using the approaches below.

    - **Test the results for special cases and extreme conditions** by substituting the special/extreme values in the equations to see if the results are physically and logically sensible. **Does your generic solution simplify to solutions for simpler special cases?**

    - **Do a dimensional analysis** of the quantities and equations in the final answer to ensure quantities and equivalences are physically sensible and meaningful.

    - **For numerical answers**, check the final value to ensure it is physically sensible and realistic. For example, a human's mass cannot be 1000 kg.

Problem-Solving Strategies in Classical Mechanics: Dimensional Analysis

Commensurable physical quantities: Quantities with the same physical units and comparable.

Incommensurable physical quantities: Quantities that are not commensurable.

The most popular physical unit system: The SI standard unit symbols:

1. time (**T**),
2. length (**L**),
3. mass (**M**),
4. electric current (**I**),
5. absolute temperature (**Θ**),
6. amount of substance (chemical amount or mol) (**N**),
7. luminous intensity (**J**).

Any other physical quantity is written in terms of the above seven fundamental SI units.

In general, the physical dimension of any physical quantity can be expressed as,

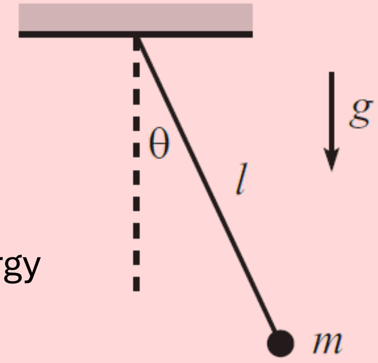
$$[Q] = \mathsf{T}^a \mathsf{L}^b \mathsf{M}^c \mathsf{I}^d \Theta^e \mathsf{N}^f \mathsf{J}^g$$

with a, b, c, d, e, f, g representing the dimension exponents.

The **first written application of dimensional analysis** has been credited to **François Daviet**, a **student of** Joseph-Louis **Lagrange**, in a **1799** article at the Turin Academy of Science.

Problem-Solving Strategies in Classical Mechanics

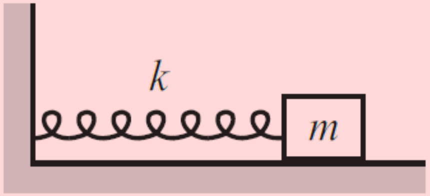
Problem 2. Dimensional Analysis.



a) A mass m hangs from a massless string of length l and swings back and forth in the plane of the paper. The acceleration due to gravity is g. What can we say about the frequency of oscillations?

b) What can we say about the total energy of the pendulum (with the potential energy measured relative to the lowest point)?

Problem 3. Dimensional Analysis.



A spring with spring constant k has a mass m on its end. The spring force is $F(x) = -kx$, where x is the displacement from the equilibrium position.
a) What can we say about the frequency of oscillations? **b)** How about energy?
c) How do the answers to parts (a) and (b) change if the Spring force equation is $F(x) = -kx + bx^2$?

Problem 4. Dimensional Analysis.

A satellite of mass m travels in a circular orbit above the earth's surface. What can we say about its speed?

Problem 5. Special Cases.

A beach ball is dropped from rest at height h.
Assume that the drag force from the air takes the form $F_d = -m\alpha v$.
The governing equations of motion are the following

$$v(t) = -\frac{g}{\alpha}\left(1 - e^{-\alpha t}\right), \quad \text{and} \quad y(t) = h - \frac{g}{\alpha}\left(t - \frac{1}{\alpha}\left(1 - e^{-\alpha t}\right)\right).$$

Prove the above equations are physically sensible for very small and large values of t compared to α.

Problem-Solving Strategies in Classical Mechanics

Problem 6. Special Cases.

A mass m with speed v approaches a stationary mass M.
The masses bounce off each other elastically.
Assume that all motion takes place in one dimension.
The final velocities of the particles are,

$$v_m = \frac{(m-M)v}{m+M}, \quad \text{and} \quad v_M = \frac{2mv}{m+M}.$$

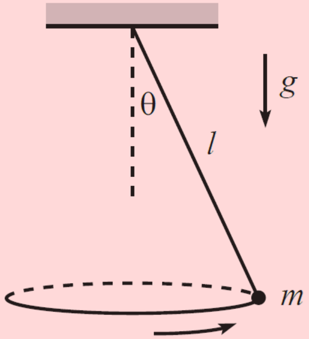
Investigate the limiting behavior of the equations for the extreme values of the two masses relative to each other ($m << M, m = M, m >> M$).

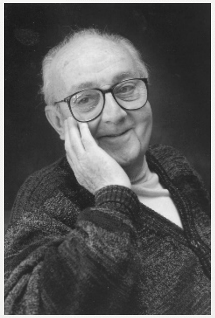
Problem 7. Special Cases.

A mass hangs from a massless string of length l.
The mass swings around in a horizontal circle, with the string making a constant angle θ with the vertical.

The angular frequency, ω, of this motion is $\omega = \sqrt{\frac{g}{l\cos\theta}}$.

Prove the behavior of the system is sensible for the extreme angle values.





All models are wrong (including physical models), but some are useful.

George E. P. Box (1919 – 2013)

Problem 8. Special Cases.

Atwood's machine

Consider the "Atwood's" machine shown in Fig. 1.7, consisting of three masses and three frictionless pulleys. It can be shown that the acceleration of m_1 is given by (just accept this):

$$a_1 = g\frac{3m_2m_3 - m_1(4m_3 + m_2)}{m_2m_3 + m_1(4m_3 + m_2)},$$

with upward taken to be positive. Find a_1 in the following special cases:

(a) $m_2 = 2m_1 = 2m_3$.
(b) m_1 much larger than both m_2 and m_3.
(c) m_1 much smaller than both m_2 and m_3.
(d) $m_2 \gg m_1 = m_3$.
(e) $m_1 = m_2 = m_3$.

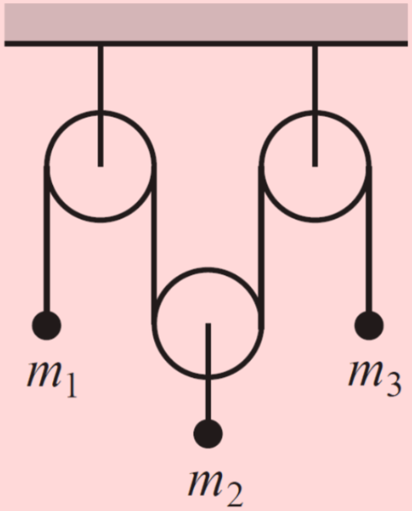


Fig. 1.7

Problem-Solving Strategies in Classical Mechanics

Problem 9. Special Cases.
If a pendulum has a period of $3s$ on the earth, what would its period be if it were placed on the moon? Use $g_M/g_E \approx 1/6$.

Problem 10. Special Cases.

Projectile with drag

Consider a projectile subject to a drag force $\mathbf{F} = -m\alpha\mathbf{v}$. If it is fired with speed v_0 at an angle θ, it can be shown that the height as a function of time is given by (just accept this here;

$$y(t) = \frac{1}{\alpha}\left(v_0 \sin\theta + \frac{g}{\alpha}\right)\left(1 - e^{-\alpha t}\right) - \frac{gt}{\alpha}. \qquad (1.19)$$

Show that this reduces to the usual projectile expression, $y(t) = (v_0 \sin\theta)t - gt^2/2$, in the limit of small α. What exactly is meant by "small α"?

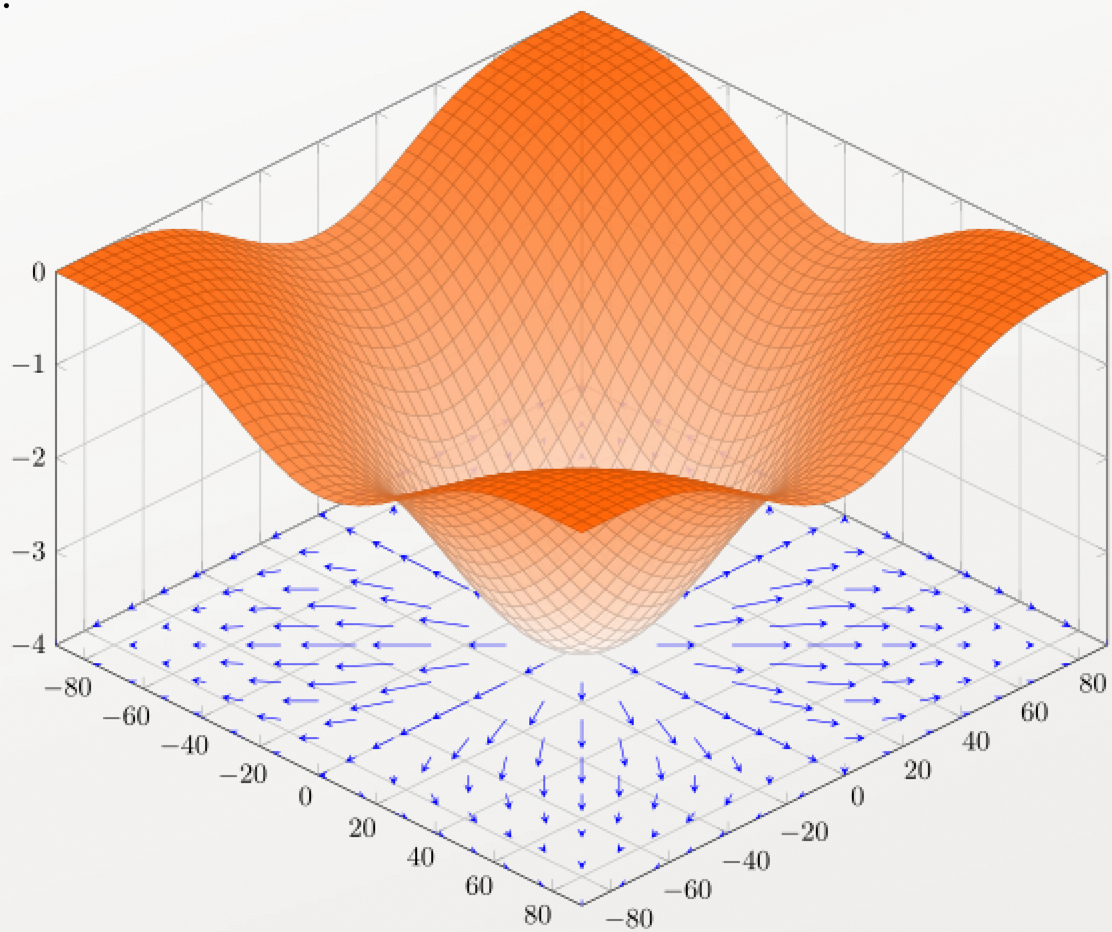
Problem 11. Matrix properties.
Show that $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$ and $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ hold.

Problem 12. Calculus. If $\boldsymbol{r}$ and $\dot{\boldsymbol{r}} = \boldsymbol{v}$ are both explicit functions of time, show that,

$$\frac{\mathrm{d}[\mathbf{r} \times (\mathbf{v} \times \mathbf{r})]}{\mathrm{d}\,t} = r^2\mathbf{a} + (\mathbf{r} \cdot \mathbf{v})\mathbf{v} - (v^2 + \mathbf{r} \cdot \mathbf{a})\mathbf{r} \ .$$

Gradient and Laplacian

Given a scalar function of coordinates $f(\boldsymbol{r})$, the **gradient** of f is defined as a vector function over the same space whose value at a given coordinate determines the magnitude of the greatest change in the value of the function f at that point, while its direction determines the direction of the greatest changes in the value of the function.



The gradient of $f(x, y) = -(\cos^2 x + \cos^2 y)^2$ depicted as a projected vector field on the bottom plane.

Gradient and Laplacian

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In Cartesian coordinates, the gradient is written as,

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k} \ ,$$

where $\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}$ are the standard unit bases of the Cartesian Coordinate system.

In spherical coordinates, the gradient is given by,

$$\nabla f(r,\theta,\varphi) = \frac{\partial f}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\mathbf{e}_\theta + \frac{1}{r\sin\theta}\frac{\partial f}{\partial \varphi}\mathbf{e}_\varphi \ ,$$

where r is the radial distance, φ is the azimuthal angle, and θ is the polar angle, and $\boldsymbol{e_r}$, $\boldsymbol{e_\theta}$, and $\boldsymbol{e_\varphi}$ are again unit basis vectors pointing in the spherical coordinate directions.

Gradient and Laplacian

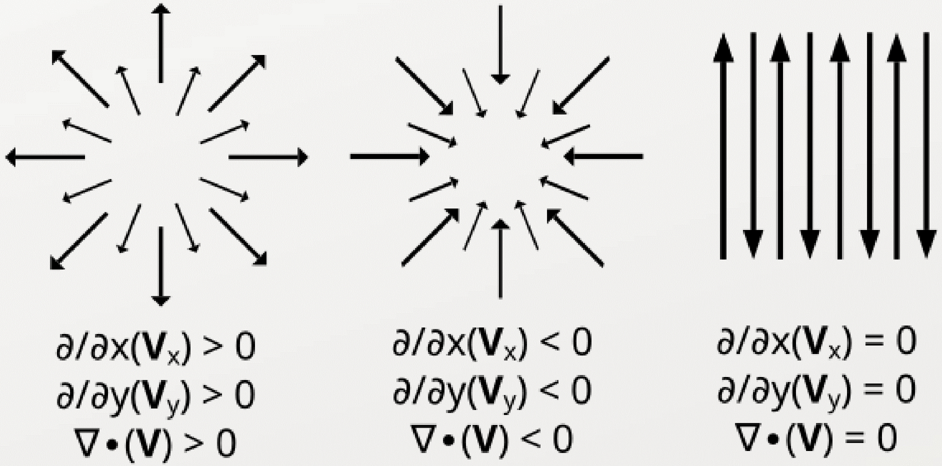
Given a scalar function of coordinates $f(\boldsymbol{r})$, the **Laplacian** of f is is defined as the divergence of the function's gradient over the same space.

In Cartesian coordinates, the Laplacian is written as,

$$\Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \ .$$

In spherical coordinates, the Laplacian is given by,

$$\Delta f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 f}{\partial \varphi^2} \ ,$$



Problem 13. Calculus. Show that,

$$\nabla(\log(|\mathbf{r}|) = \frac{\mathbf{r}}{r^2} \ .$$

Problem 14. Calculus. Show that,

$$\nabla^2(\log(|\mathbf{r}|) = \frac{1}{r^2} \ .$$

Problem 15. Dimensional Analysis.

Damping

A particle with mass m and initial speed V is subject to a velocity-dependent damping force of the form bv^n.

(a) For $n = 0, 1, 2, \ldots$, determine how the stopping time depends on m, V, and b.

(b) For $n = 0, 1, 2, \ldots$, determine how the stopping distance depends on m, V, and b.

Be careful! See if your answers make sense. Dimensional analysis gives the answer only up to a numerical factor. This is a tricky problem, so don't let it discourage you from using dimensional analysis. Most applications of dimensional analysis are quite straightforward.